Use Case 3 Student Guide: Grid Congestion Prices

The challenge

Where and when do congestion-related electricity price impacts occur, and how do those impacts compare across locations?

New York's wholesale electricity price is not the same everywhere. When transmission lines between regions run at their limits, cheaper power cannot reach where it is needed. Prices then rise on the far side of the bottleneck. The part of the price caused by these limits is the **congestion component**.

Your team gets three years of real-time congestion price data for New York's 11 pricing zones, plus power flows across the Moses South interface, from the New York Independent System Operator (NYISO). You will compare zones and time periods, find patterns, and explain them.

Why it matters: congestion costs show where the grid needs more transmission capacity. Projects such as NYPA's Smart Path Connect in northern New York and the Champlain Hudson Power Express (CHPE) line into New York City aim to relieve these limits.

A successful team will:

- Compare congestion price components across zones or time periods
- Show calculations that identify differences or patterns
- Present at least one clear chart
- Explain what the results suggest, and state what **cannot** be concluded from this data

Choose your level

Every team must finish the Minimum deliverable first. Move up only once it works.

Level	Project	You must deliver	Best data file
Minimum	Congestion price comparison	1) Identify relationship between LBMP and Congestion costs and present it in a chart; 2) Compare avg LBMP avg cross zones and find the top three zones with highest LBMP 3) Compare avg Congestion costs cross zones and find out top three zones with heaviest negative congestion costs	Monthly congestion by zone
Intermediate	Congestion pattern analyzer	-Map out time-series based congestion costs for the top 3 zones in a chart	Daily + hourly

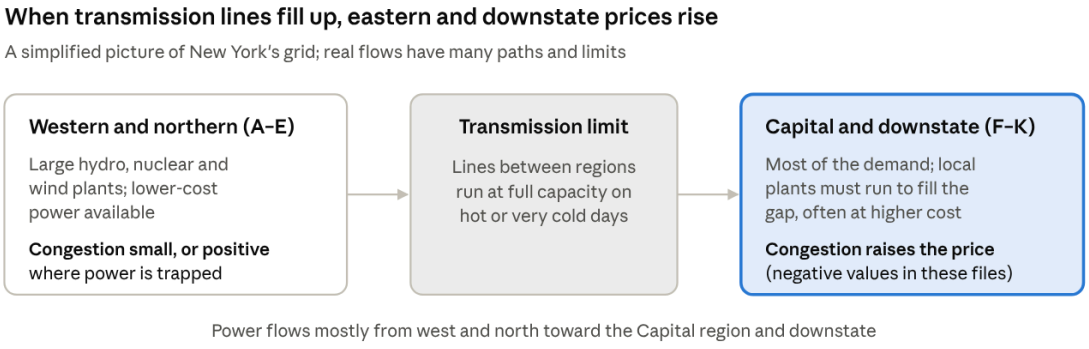
Level	Project	You must deliver	Best data file
		- Any findings on variation on seasonality, monthly hourly, weekend vs weekday, etc. Demo it in chart.	profile+monthly profile
Advanced	Congestion mapping and connect with Smart Path flow (Moses South)	- Trend analysis on Moses South Flow from 2023-2025 -Explore correlation between Moses South flow and congest cost for Zone D, F, J.	Mose south flow daily file +Congest daily file

Key ideas

Every hour, NYISO sets a price for each zone called the **LBMP** (Location-Based Marginal Price), in dollars per megawatt-hour ($/MWh). It has three parts:

LBMP = energy component + losses component - congestion component

- **Energy:** the cost of the next unit of power, the same in every zone in a given hour
- **Losses:** a small adjustment for power lost as heat on long lines
- **Congestion:** the extra cost when transmission limits stop cheaper power from reaching a zone. It is usually zero, negative where power is hard to deliver (the more negative, the heavier the congestion) and positive where power is trapped.



concept diagram · how congestion arises

Smart Path Connect (NYPA)

A major upgrade of transmission lines from **Massena** → **Croghan** → **Marcy** → **Central NY**, including high-voltage- rebuilds and capacity increases.
Purpose:

- Move more renewable and hydropower out of the North Country

- Reduce internal bottlenecks between **Zone D → Zone E → Zone F**
- In the interface flow dataset, Moses South contains the flow from Smart path connect

New York is split into 11 pricing zones, labeled A to K from west to east and then south:

Zone	Name	Region	Representative city
A	West	Upstate	Buffalo
B	Genesee	Upstate	Rochester
C	Central	Upstate	Syracuse
D	North	Upstate	Plattsburgh
E	Mohawk Valley	Upstate	Utica
F	Capital	Upstate	Albany
G	Hudson Valley	Lower Hudson Valley	Poughkeepsie
H	Millwood	Lower Hudson Valley	Mount Kisco
I	Dunwoodie	Lower Hudson Valley	Yonkers
J	New York City	New York City	Manhattan
K	Long Island	Long Island	Hauppauge

The representative city is a reference point for mapping, not the zone's boundary.

The dataset

You get NYISO's real-time zonal prices (5-minute intervals) and Moses South interface flows from Jan 1 2023 to Dec 31 2025, averaged to hourly, daily and monthly values so they open in any spreadsheet. The hourly price file covers 26,276 hours for all 11 zones.

File	Rows	One row is	Use it for
`Congestion_monthly_by_zone_2023_2025.csv`	396	one zone in one month including both avg congestion costs and LBMP	Comparing zones and months (start here)
`Congestion_Daily_by_zone_2023_2025.csv`	1,096	one day, all 11 zones	Seasons, weekdays vs weekends, extreme days

File	Rows	One row is	Use it for
`Congestion_Hourly_by_Month_2023_2025.csv`	864	one hour of an average day in a month	When during the day congestion appears
`Congestion_Raw_2023_2025.csv`	26,276	hourly based raw congest cost for all zones	big file for more detailed time series analysis
`Mosesouth_flow_2023_2025_raw.csv`	26,296	Hourly based moses south flow	A big file for more detailed time series analysis
`Mosesouth_flow_2023_2025_dailyavg.csv`	1,096	Daily avg flow	For map of daily trend and compared it with daily congestion costs
`zones_reference.csv`	11	one zone	Names, regions and map coordinates

The full column list is in `Data_Dictionary.docx`.

Units. All prices are in **$/MWh**, dollars per megawatt-hour of electricity. For scale, 1 MWh is roughly what an average home uses in one to two months. Moses South flows are in MW, averaged over the hour or day.

Sign convention. In these files, a **negative** congestion value means congestion **raised** the price in that zone, because LBMP = energy + losses − congestion. A positive value means congestion lowered the price, typically where power is trapped.

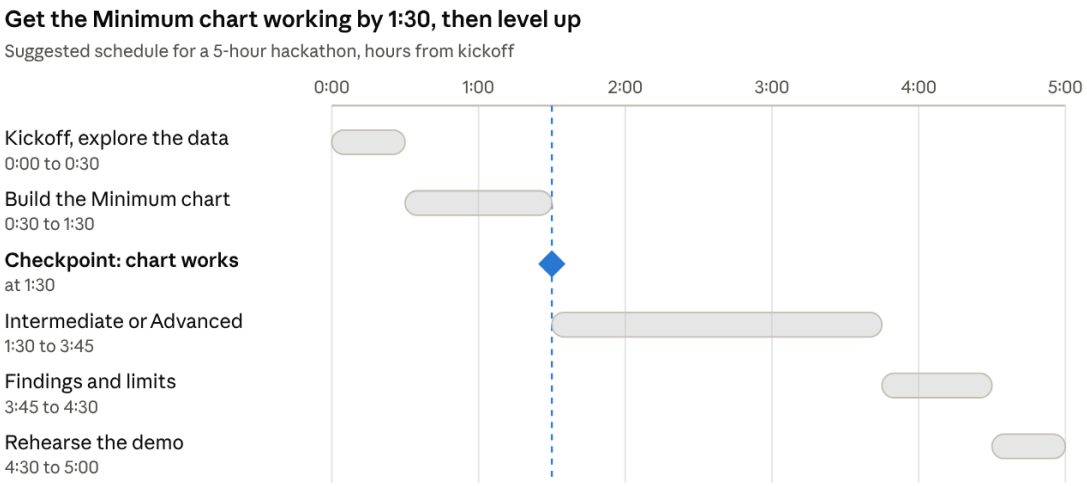
Averages hide spikes. A monthly average of −$5/MWh can contain a few 5-minute intervals below −$100/MWh. Use the min and max columns, the daily file or the hourly raw file to find them.

Know the limits before you present.

- These are **Real-time** prices, which has high volatility and can be impacted by many factors
- Prices are per MWh. Without knowing how much power each zone used, you cannot turn them into total dollars of congestion cost.
- The data shows **where and when** congestion happened, not which transmission line caused it.
- Congestion depends on weather, fuel prices, demand, outages and new projects all at once. A change between years cannot be pinned on one cause from this data alone.

Your five hours

Have a working Minimum chart by the 1:30 checkpoint, then spend the middle of the day on deeper patterns.



suggested schedule · 5 hours, 1 checkpoint

Split the work so nobody waits on anyone else:

Role	Owns
Data lead	Loading the files, filtering zones and periods, checking signs and units
Visualization lead	Charts, labels, colors, filters or map
Analyst	Finding the story: which zones, which seasons, which hours, and why; drafting the "what cannot be concluded" statement
Presenter	The 3-minute demo script and timekeeping; joins whichever role is behind

Teams of three can merge Analyst and Presenter.

How to build each level

Your team chooses the tools. We provide the data; how you analyze and present it is up to you. Spreadsheets (Excel, Google Sheets), Python or R, BI tools (Power BI, Tableau, Looker Studio) and mapping tools can all reach the Advanced level. Pick what your team already knows.

Minimum: congestion price comparison

1. Open `Congestion_monthly_by_zone_2023_2025.csv`.
2. Chart avg_lbmp against avg_congestion_cost for each zone and month (a scatter plot works or any right plot works) and describe the relationship.
3. Calculate each zone's average LBMP for the period you choose, such as one year or all three, and find the three zones with the highest LBMP.
4. Calculate each zone's average congestion cost the same way and find the three zones with the heaviest (most negative) congestion costs.

5. Make a bar chart of each ranking, ordered west to east (A to K) or from highest to lowest.
6. Write your explanation and your "what cannot be concluded" statement.

Intermediate: congestion pattern analyzer

Answer **when** as well as **where**. Ideas:

- A time series of congestion cost for your top 3 zones across all 36 months (daily or monthly file)
- Average congestion by hour of day in a summer month vs a winter month (hourly by month file)
- The 10 most congested days for one zone, and what they have in common (daily file)
- Weekdays vs weekends: compare average congestion cost using is_weekend (daily file)
- Seasons: compare winter, spring, summer and fall using the season column (monthly file)

Advanced: congestion mapping and Smart Path flow (Moses South)

Use the Moses South flow files together with the congestion files:

- **Trend:** chart daily Moses South flow from 2023 to 2025 (Mosesouth_flow_2023_2025_dailyavg.csv). How did the average flow change from year to year?
- **Correlation:** join the daily flow file to Congestion_Daily_by_zone_2023_2025.csv on date. Calculate the correlation between flow and congestion cost for zones D, F and J, and show each as a scatter plot. You can repeat it hour by hour with the two raw files. A correlation shows the two move together, not that one causes the other.
- **Map (optional):** place each zone at its coordinates from zones_reference.csv and size or color the markers by average congestion cost.

Demo and judging

You have about three minutes. Cover, in order: your question, the data you used, your chart (live, not a screenshot), your findings, and your "what cannot be concluded" statement.

The "what cannot be concluded" statement is part of the Minimum deliverable. Name at least one thing your audience might assume from your chart that the data does not prove. For example:

- Higher congestion in a zone does not tell us which transmission line was the bottleneck.
- A drop in congestion from one year to the next does not prove that a specific project caused it.
- Average $/MWh does not tell us total dollars paid, because we do not have each zone's usage.

Criterion	Weight	What judges look for
Data & Technical Execution	30%	Accuracy of analysis, soundness of methodology, appropriateness of assumptions and overall technical rigor.
Visualization & Insights	20%	Clarity and effectiveness of visualizations, whether key insights are immediately apparent, appropriate chart selection, dashboard usability, and ability to answer the business question at a glance.

Criterion	Weight	What judges look for
Innovation, AI & Automation	15%	Creative use of analytics techniques, AI/GenAI capabilities, machine learning, automation of data ingestion or processing, and innovative approaches to solving the problem.
Potential Business Impact	15%	Practical value of the solution, applicability to real-world challenges, scalability, or decision-making impact.
Demo & Storytelling	20%	Clear and compelling presentation, logical flow, effective communication of findings, ability to explain technical concepts, staying within time limits, and engaging delivery

Before you present, check:

- ☐ Congestion compared across zones or periods
- ☐ Calculations shown
- ☐ At least one chart
- ☐ A short explanation of what the results suggest
- ☐ A "what cannot be concluded" statement
- ☐ Data source and time period stated on the chart or slide

Questions to explore

Each of these has a real answer in the data.

- Which zone had the heaviest (most negative) average congestion cost in each year? Is it the same zone every year?
- Is congestion worse in winter or in summer? Does the answer differ by zone?
- Zone D (North) often shows positive congestion cost. What does a positive value mean for the price there?
- Look at early February 2023 in the daily and hourly raw files. What happened across the zones?
- At what hour of the day does New York City congestion peak in July? Is it the same in January?
- How did 2024 compare with 2023 and 2025? Was the change the same in every zone?

Common mistakes

- **Getting the sign backwards.** In these files a negative congestion cost means congestion raised the price, and a positive one means it lowered it.
- **Averaging monthly averages without weights.** February has fewer hours than March; weight by the number of hours in each month, or average the hourly raw file directly.
- **Mixing up components.** Congestion is one part of the LBMP, not the whole price.
- **Turning $/MWh into total dollars.** You would need each zone's MWh of usage, which is not in this dataset.
- **Claiming a cause.** The data shows patterns. Explain likely reasons, but say what you cannot prove.

Sources

- NYISO real-time market zonal LBMP, report P-24A: the 5-minute interval data on LBMP price
- NYISO Interface flow data: 5-minute interval data on interface flow